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## Introduction

Type 2 diabetes mellitus (T2DM) is a chronic, non-communicable metabolic disease characterized by elevated blood glucose levels resulting from insulin resistance and impaired insulin secretion. Unlike type 1 diabetes, which is autoimmune in nature, type 2 diabetes develops gradually and is strongly influenced by lifestyle, environmental, and genetic factors. The disease has become one of the most pressing public health concerns due to its increasing prevalence and its association with serious long-term health complications.

Globally, type 2 diabetes affects hundreds of millions of people and continues to rise at an alarming rate. In the United States, approximately 90–95% of all diabetes cases are classified as type 2 diabetes. Incidence rates have increased across nearly all age groups, including adolescents and young adults, largely due to rising obesity rates, sedentary lifestyles, and poor dietary habits. Certain populations, including racial and ethnic minority groups and individuals of lower socioeconomic status, experience disproportionately higher prevalence and worse health outcomes.

Mortality associated with type 2 diabetes is often indirect. While diabetes itself may not be listed as the primary cause of death, it significantly increases the risk of cardiovascular disease, stroke, kidney failure, and infections, all of which contribute to premature mortality. In addition to mortality, the disease leads to substantial disability, including vision loss, neuropathy, and lower-extremity amputations. The economic burden of diabetes is also considerable, with billions of dollars spent annually on medical care, medications, and treatment of complications, as well as indirect costs related to lost productivity.

Understanding the etiology, pathophysiology, and risk factors of type 2 diabetes is essential for effective public health planning and clinical practice. This knowledge supports early intervention, targeted prevention strategies, and improved disease management to reduce complications and improve overall health outcomes.

## **Etiology**

The etiology of type 2 diabetes mellitus is multifactorial and involves an interaction between genetic predisposition and environmental and lifestyle factors. Genetics play an important role, as individuals with a family history of diabetes are at a higher risk of developing the disease. Several genetic variants have been identified that influence insulin sensitivity, glucose metabolism, and pancreatic beta-cell function.

Lifestyle and environmental factors are the primary contributors to disease onset. Diets high in refined carbohydrates, added sugars, and saturated fats promote weight gain and insulin resistance. Physical inactivity further worsens insulin sensitivity by reducing glucose uptake in skeletal muscle. Over time, these behaviors contribute to metabolic dysfunction and elevated blood glucose levels.

Obesity, particularly abdominal or visceral obesity, is one of the strongest etiological factors for type 2 diabetes. Excess adipose tissue releases inflammatory substances that interfere with insulin signaling, leading to chronic low-grade inflammation and insulin resistance. Additional contributing factors include aging, chronic stress, inadequate sleep, and socioeconomic conditions that limit access to healthy foods and preventive healthcare. Together, these factors disrupt normal glucose regulation and lead to the development of type 2 diabetes.

## **Pathophysiology**

The pathophysiology of type 2 diabetes mellitus is primarily characterized by insulin resistance and progressive pancreatic beta-cell dysfunction. At the cellular level, insulin resistance occurs when muscle, liver, and adipose cells fail to respond adequately to insulin. As a result, glucose uptake by muscle cells is reduced, and the liver continues to produce glucose despite elevated blood sugar levels.

In the early stages of the disease, pancreatic beta cells attempt to compensate for insulin resistance by increasing insulin secretion. This compensatory mechanism may maintain near-normal glucose levels temporarily. However, chronic metabolic stress, inflammation, and elevated fatty acid levels eventually impair beta-cell function, leading to decreased insulin production and persistent hyperglycemia.

Prolonged elevated blood glucose causes damage at the tissue and organ levels. Microvascular complications include diabetic retinopathy, nephropathy, and neuropathy, resulting from damage to small blood vessels. Macrovascular complications involve large blood vessels and increase the risk of coronary artery disease, stroke, and peripheral arterial disease. These complications account for much of the morbidity and mortality associated with type 2 diabetes.

## **Risk Factors**

Type 2 diabetes is associated with both non-modifiable and modifiable risk factors.

Non-modifiable risk factors include genetic predisposition, advancing age, and certain racial and ethnic backgrounds that experience higher disease prevalence. Individuals with a family history of diabetes are at a significantly increased risk.

Modifiable risk factors include obesity, physical inactivity, unhealthy dietary patterns, smoking, and excessive alcohol consumption. Poor sleep quality and chronic stress also contribute to hormonal imbalances that negatively affect glucose regulation. Socioeconomic factors, such as limited access to healthcare, low income, and reduced health literacy, further increase disease risk.

These risk factors often interact and reinforce one another. For example, socioeconomic disadvantage may limit access to nutritious foods and safe exercise environments, increasing obesity and insulin resistance.

Three major challenges in preventing type 2 diabetes include:

1. Sustaining long-term behavior change related to diet and physical activity.
2. Addressing health inequities that limit access to preventive resources and early intervention.
3. Delayed diagnosis due to the asymptomatic nature of early-stage disease.

## **Prevention Strategies**

Prevention strategies for type 2 diabetes focus on lifestyle modification, early detection, and effective disease management. Primary prevention aims to reduce disease incidence through healthy eating, regular physical activity, and weight management. Community-based programs and public health campaigns play a key role in promoting these behaviors.

Secondary prevention emphasizes early screening and identification of individuals with prediabetes or undiagnosed diabetes. Early intervention can delay or prevent disease progression through lifestyle changes and clinical monitoring. Tertiary prevention focuses on managing blood glucose levels, preventing complications, and improving quality of life through medication, self-management education, and regular medical care.

Prevention strategies for non-communicable diseases like type 2 diabetes differ from those for communicable diseases. Communicable disease prevention often relies on vaccination, infection control, and interrupting transmission. In contrast, diabetes prevention requires sustained behavior change, environmental support, and policy-level interventions that address social determinants of health.

### **Public Health Implications**

Type 2 diabetes has a significant impact on public health systems, populations, and economies. The disease places a heavy burden on healthcare systems due to long-term treatment needs and costly complications. On a population level, diabetes contributes to reduced productivity, increased disability, and higher rates of premature mortality.

Economically, the disease results in substantial direct medical costs and indirect costs such as lost wages and caregiver burden. Addressing type 2 diabetes requires coordinated public

health efforts that integrate clinical care, community interventions, and policies aimed at reducing risk factors and promoting health equity.

## **Conclusion**

Continued research into the etiology, pathophysiology, and risk factors of type 2 diabetes mellitus is essential for improving prevention and treatment strategies. As the prevalence of the disease continues to rise, evidence-based interventions that address biological, behavioral, and social determinants of health are critical. Strengthening prevention efforts, expanding access to care, and promoting health equity will be key to reducing the long-term burden of type 2 diabetes and improving population health outcomes.

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